COMP0218: Industrial Project in Robotics and AI

Individual Report:
Automating Battery Manipulation with VLA Models

Group 04 - Team Extensify

Aayan Islam — 24102520

Department of Computer Science
University College London
29/05/2026

Contents

1 Individual Contributions

During the project's decision phase, I took an active role in planning and feasibility analysis. I helped decide which company would be most beneficial to the team by finding pros and cons regarding each company and comparing them between all the choices as well as matching them to our teams desires and technical capabilities. I advocated for the Extend Robotics VLA project, identifying it as a project which would be best suited for our team. This decision was made by looking at each company had to offer as well as location and ability to receive support from industry experts. Extend Robotics proved to align with all of these and was within our capacity to develop and test a minimum viable product within the time frame given.

1	Company	Project	Technical skills	Location	Project score	Company score	Location score	Presentation score	Total
2	Dobot	imitation learning with 2 robot arms	reinforcement learning, isaac sim	remote	6	3	5	3	17
3	Extend robotics	work on haptic torque-aware VLA models	ros2, cuda, pytorch, VLA, robot manipulation, reinforcement learning	in office (Reading?)	8	6.5	7	8.5	30
4	Laelaps	multi-robot coordination and navigation	python, ros2, linux, multi-agent systems, object detection, SLAM	remote (can work in Zurich)	7	5	3	5	20
5	Mathworks	Adaptive Palletizing with Simulation	matlab, simulink, control	remote					0
6		Human Motion Recognition Using IMUs	matlab, simulink, deep learning, using IMUs, sensor fusion	remote					0
7		Sensor Fusion for Autonomous Systems	matlab, simulink, sensor fusion	remote					0
8		Trajectory Planning for Multirotor Drones	matlab, simulink, control	remote					0
9		Simulation-Based Design of Humanoid Robots	matlab, simulink, control	remote					0
10	Cisco	autonomous robots with embodied AI	python, multi-agent systems, agentic AI, computer vision, networking	in office (London?)	4	9	9	7	29
11	Knight Frank	drones for biodiversity	python, computer vision, image segmentation, deep learning, drones(?)	in office (where?)	6.5	8	10	7.5	32
12		identification of property/sale images to improve satisfaction	python, computer vision, machine learning	in office	6				
13									

Figure 1: Scoring the different companies in an attempt to decide which one is most beneficial

During the project itself, I was tasked with data pre-processing. This involved taking all the sample footage clips and cropping all of them such that all the samples ended under the same conditions. Furthermore, using the LeRobot visualiser, I ensured that my split of the data samples were correct by checking the action and state plots of all the joints as well as checking the boolean end values. By doing this, any invalid data samples were removed to ensure the data set was as clean as possible for training. To remove local hardware bottlenecks, I configured and provided a remote GPU workstation. This allowed the team to train the models which was then used on the arm itself.

On site, I worked with my other team members to collect data needed for training. By using the Meta Quest 3, I was able to use a digital twin of the xArm to control the physical arm to carry out the task of picking up a battery and dropping it into the slot. This was done to generate a high number of contact-rich manipulation demonstrations. Towards the end of the project, I was also tasked with using the force-feedback sensors on the gripper to push down a button. By gathering these samples, I was able to help the company generate more data regarding force feedback which they can then use into developing new models in the future.

Figure 2: Gathering tele-operation data samples at the Extend Robotics facility.

Figure 3: Testing out the force feedback sensor on the gripper at the Extend Robotics facility

At the end of our project, I was heavily involved in the evaluation of the project. I took responsibility for writing up what the team thought went well with the project as well as what we had found to be of concern and could be improved. By critically assessing this process, I was able to identify core logistical and technical setbacks—such as hardware integration, sample size constraints and extract key benefits from our approach. Additionally, I highlighted specific, actionable upgrades such as multi-view camera integration altering the motion of the grasp.

Within our final presentation, I was in-charge of creating the slides of our final conclusion. This involved talking about what we as a team found went well with the project, what challenges we faced as well as future improvements/works we could possible implement. During the presentation itself, I presented slides on the data collection for both our model iterations and our training and evaluations for our V2 model. I was also able to present all the conclusive findings I had collated together. Overall, I believe the presentation went very well.

2 Critical Reflection

2.1 Reflection on learning experience

Engaging with an industry-led project with Extend Robotics has fundamentally shaped my understanding of how artificial intelligence is applied within the industry. Before this project, I had never worked with a VLA or even any AI within the industry. Typically within university, data given to us clean and compatible. However, when gathering data with the xArm, I was shown the true reality of data collection and data processing. There were lots of inconsistencies in the data as well as legacy systems. This lead us to move away from using a gRPC framework to a WebSocket protocol. This has taught me a valuable lesson in how AI deployment is not a simple systematic path but rather full of fallbacks and re-attempts. It relies heavily on engineering and flexibility such as removing extra objects, keeping camera views constant and maintaining smooth motion. All of these are necessary to replicate human motion. There is also a large requirement for time commitment and dedication as VLAs proved to be long to train and need many iterations to produce a robust model.

2.2 Personal development and skills acquired

From a technical perspective, this project has shown that there is a difficult learning curve with VLAs. By being able to work on data collection, I was able to carry out imitation learning on a digital twin which increased my competency with the meta quest 3 mapping human physical movements into the actual arm through the digital arm. Furthermore, being able to process data has allowed me to learn how to augment data to ensure efficiency. For the first time, I also learned how to use the UCL remote GPU system by connecting through a VPN and then accessing the remote GPU system.

I have also improved my soft-skills while carrying out my project at Extend Robotics. Being able to communicate with the team regarding VLAs has not only improved my knowledge of VLAs but also benefited my communication skills. Additionally, we carried out presentations to academic leaders and industry supervisors which not only allowed me to work on my presentation skills but also the ability to condense information into slides and be able to work as a team to present within a time frame.

2.3 Reflection on teamwork dynamic

I personally believe that the team worked extremely well and effectively. We were able to have good discussions and even when there was opposing ideas, we would all listen to the points provided each other to come to an overall agreement. We would systematically progress through any problems. When planning our trips, we would simply have polls to effectively see who was able to attend and who couldn't. This made it very easy to know how many people were coming so that we can let the Extend team know in advanced how many people they can be expecting.

To ensure that all members could work on the data together, a GitHub repository was set up. This would enable all members to work on the datasets and upload the edits and augmentations. By doing this sequentially, we prevented any kind of conflict. We had also decided beforehand which parts of the dataset each member would work on which helped keep everything smooth.

In general, the team had a lot of fun working on the project and enjoyed working with each other. I believe that due to the positive environment provided by our team, it made progression much quicker.

3 Peer Assessment (IPAC)

Team Member	Contribution to the Project	Quality of Work	Team Collaboration	Initiative and Problem Solving	Time Management
Yash	5	5	5	5	5
Helitha	5	4	5	5	5
Aryan	4	4	5	5	4
Andrew	4	4	5	4	4
Aayan (Myself)	4	3	5	3	3

Table 1: Peer Evaluation Matrix

3.1 Justification for scores

3.1.1 Yash

Yash was our teams project manager. He took it upon himself to lead the team as well as ensure strong communications with the Extend Robotics team. He planned all of the trips to Reading and ensured all team members were aware of the dates as well as creating instruction sheets on how to pre-process data and set up the remote GPU workstations. He also kept us all up to date as soon as new information was given to him. Even when the progression of the project was slow, he made sure to show resilience and a positive attitude to help keep the team motivated. Therefore, to justify that, I have given him a score of 5 for contribution and team collaboration. Yash also worked on the inference pipeline that allowed our VLA models to communicate with the xArm itself and enable us to use the arm autonomously which is why I have decided to give a score of 5 for quality of work. To further reinforce this, Yash also handled the GitHub repository allowing all team members to work on the dataset. Yash attended every trip to Reading and provided valuable resources and communication all whilst having to balance his day to day tasks. For that reason, I have also given him 5 for initiative and time management.

3.1.2 Helitha

Helitha took charge of the simulation which allowed us to test the kinematics of the arm as well as carry out testing without having to work on the physical xArm. The development of the simulation was of high quality and also took him lots of time and dedication. Helitha also developed a pipeline to utilise GR00T which was another VLA model which had around 3B parameters. Additionally, Helitha maintained good communications with the staff at Extend Robotics, further creating a positive atmosphere for everyone as well as providing valuable insights into the xArm and VLA models. Helitha also made sure to include everyone when working on different task, explaining them as well as showing demos to allow the rest of the team to understand the progression. These reasons are why I have decided to give Helitha a score of 5 for contributions, team collaboration, initiative and time management. The justification for giving Helitha a 4 for quality is due to the fact that the simulation was not used as frequently as originally expected. We had been told by the Extend team that there would be an extremely large sim-to-real gap which would lead to the real arm performing differently to what the simulation showed. Instead, the simulation was used to prove tele-operation can be done. Therefore, I have given Heltiha a score of 4 for the quality of work.

3.1.3 Aryan

Aryan handled the majority of the data pre-processing. I believe he was one of the most proactive members of the team. When we were sent the data set and needed to process the data, he would be one of the first team members to work on it and complete the task given. He has also provided valuable insights and ideas towards the development of the pipeline and how to fine tune the VLA models. Therefore, I have given him a score of 5 for team collaboration and initiative. Aryan also brought forth the concept of chattering for the xHand due to latency and signal strength. Originally, this was seen as an idea with not enough proof to support it. However, during our time at the Extend Robotics facilities, we were able to see first hand the xHand thumb joint start chattering even though the person wearing the motion glove was completely still. However, when compared to how much was done by Yash and Helitha, I personally believe that slightly less was done, hence my score rating of 4 for the other categories.

3.1.4 Andrew

Andrew was in charge of working on the arm itself. However as we were no longer working with both the xArm and xHand, his tasks were merged into general data collection. Similarly, Andrew has contributed a great amount of work, however, when compared to Yash and Helitha, he has done less than them. Therefore, I have given a contribution score of 4. Andrew would be relatively fast to complete any tasks given and would also offer to help others. This would help when a team member was unable to carry out the processing tasks as soon as possible. The episodes that he worked on were of high quality and he would justify in detail any episodes that were not valid and should not be included in the final training set. I have also decided to give him a score of 4 for initiative as he would start the work that needed to be done relatively early showcasing that he can take the initiative. Hence, I have given him a score of 5 for team collaboration and 4 for quality of work. The reason I did not give a 5 is because I believe other members have completed a greater amount of work compared to him in terms of the pipeline and inferencing.

3.1.5 Aayan (Myself)

I was tasked with originally working on the Xhand. Due to the complexity and limited scope and time frame, I instead started working with the 2 finger gripper. My main contributions were during the initial decision phase as well as the final evaluations. Within the project, I mostly was given tasks by my project manager and completed them. These tasks involved pre-processing and gathering data samples for the grasping task as well as the force feedback task. I did aid in providing a remote GPU to enable for training. I believe I provided useful insights into training as well as inference development which is why I believe my contribution is rated as a 4 and collaboration as 5. As for the rest of the sections, I have rated myself with a score of 3. Whilst I ensured my work was of high standard, I feel that I had a lower impact of intuition and taking the lead with any issues that were picked up. It was more so that other ideas were pitched and I would use them to form my own ideas and conclusions. Furthermore, as the trips to the Extend facilities were decided typically the day before, it became very difficult for me to be able to travel. This was not only due to my long travel times which was around 2.5 hours but also my part time work which I was unable to cancel the day before. Therefore, I believe my score of 3 for time management is justified as I tried to help as much as possible but it was much less compared to other members of the team.

3.2 Team Dynamics

Overall, I believe that the team maintained good communications with each other. Each member worked proactively and always had always strove to do the best they can. If one team member was unable to do a specific task, they would make sure the rest of the team was aware. They would also take it upon themselves to try and do extra after to help progression. Additionally, the teamed aimed to split work as equally among all members. During pre-processing, each team member was given a split of the data to work on.

4 Conclusion

4.1 Overall Personal Achievements

Completing this industry-led project marks a significant milestone in my growth and development as a robotics and AI engineer. Over the past five months, I have been part of data gathering, utilising a Meta Quest 3 to tele-operate and interact with a digital twin of the xArm using Extend Robotics AMAS software application. Beyond raw data gathering, I have also been part of processing the data, ensuring the recorded episodes were viable to be used within training . To ensure that there were no hardware bottlenecks, I set up and configured a remote GPU workstation to allow for multiple models to be trained at once. I had the extra accomplishment of being the only member of the team to experiment with and evaluate the physical force feedback mechanism in the two finger gripper. This has provided me with insight into tactile data for future integration. Successfully managing all of these deliverables whilst also balancing other academic and non-academic commitments under strict timelines has thoroughly demonstrated my capacity to operate within an industry standard environment to deliver robust, high quality results.

4.2 Key Insights Gained

This project has provided me with profound exposure to Vision-Language-Action (VLA) architectures, offering insight into the real world deployment and practical applications of embodied AI. Working systematically through each stage of development has revealed that optimising a VLA policy extends far beyond just increasing the number of episodes for training. True model robustness comes from consistency, whether it stems from removing unnecessary items in the image frame, or having the same team member carry out the task to ensure the model can learn the move style of that one specific person. I've also discovered that by ensuring the operational speed of the arm remains at a steady speed, the xArm is more likely to move smoothly rather than jump around. All of these insights have shown me how to optimally train a model to replicate human motion.

Furthermore, I learned that deploying VLAs onto physical hardware requires a high emphasis on system compatibility. These were simply different communication protocols, such as the gRPC framework compared to the Websocket protocol. Integrating rapidly changing sensor data with continuous robotic joint trajectories presents a small bottleneck which must be resolved using synchronization and strategic pre-processing rather than just brute-force modeling and training with as many episodes as possible. Working with different models has deepened my understanding of how model parameter capacity and inference latency can affect the accuracy and robustness of a model. Ultimately, this experience has changed my perspective of VLAs from just being another AI that needs training, to viewing them as a highly integrated software engineered for robust and reliable deployment.